

An aerial photograph of a city, likely Zurich, showing a river (Limmat) flowing through the center. The river is surrounded by dense urban development, including historic buildings and modern structures. A large blue semi-transparent rectangle is overlaid on the left side of the image, containing the title and authors. The bottom of the image shows more buildings and greenery.

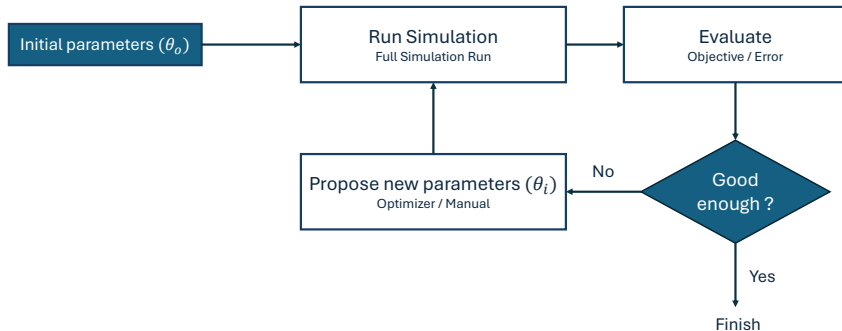
Online Calibration of MATSim Models via Stochastic Batch Optimization

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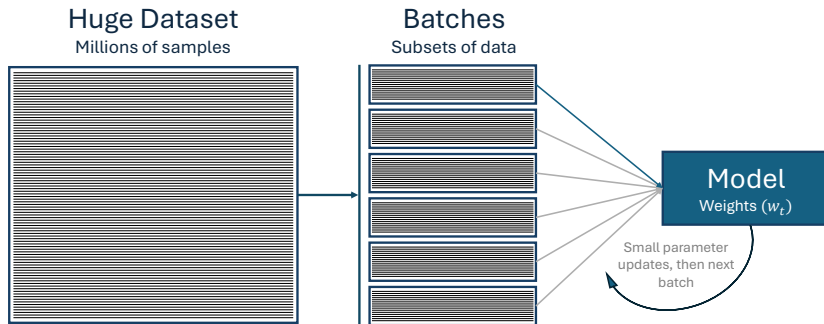
MATSim User Meeting 2026, Paris

Still tuning, running, checking, repeating?

We know the routine: adjust parameters, run MATSim, check mode shares.



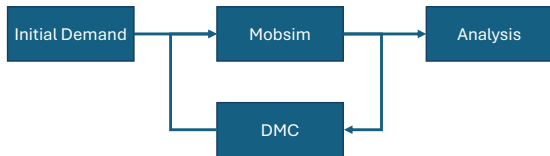
Meanwhile, in machine learning ...



$$w_{t+1} = w_t - \gamma_t \nabla_w \mathcal{L}(w_t; \mathcal{B}_t)$$

Small batches. Frequent updates. No full-dataset gradient required.

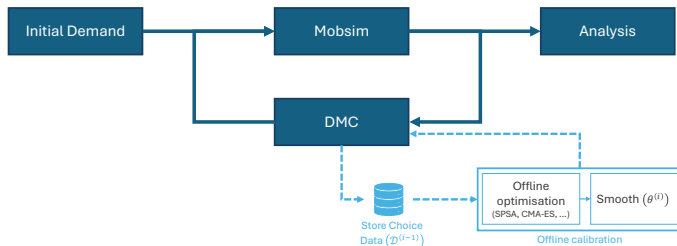
The DMC loop already gives us a batch



- In our eqasim/DMC setup, about **5% of agents replan** each iteration.
- They evaluate available modes using current travel times, costs and utility parameters.
- Their choice situations become **one calibration batch**.

The full population still runs through Mobsim.

Put calibration inside the iteration loop



1. **Log** choice variables for the replanning agents.
2. **Optimize** parameters by replaying their choices against observed shares.
3. **Smooth and apply** the update; continue the same MATSim run.

Key assumption: choice variables change little between consecutive iterations.

Learn from each batch without chasing its noise

$$\underbrace{\theta^{(i)}}_{\text{updated parameters}} = \gamma_i \underbrace{\theta^{(i-1)}}_{\text{previous parameters}} + (1 - \gamma_i) \underbrace{\theta^{*(i)}}_{\text{batch proposal}}$$

- Obtain a batch proposal **without rerunning Mobsim**.
- Start with larger adjustments; increase γ_i toward 1 to stabilize.
- Use each batch to find an update direction as network conditions evolve.

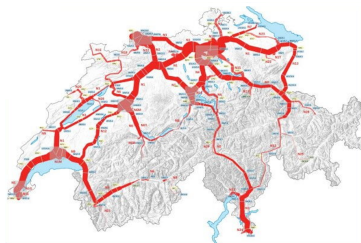
Parameters and travel choices **co-evolve in one run**.

Experiment: Different cantons, different travel preferences

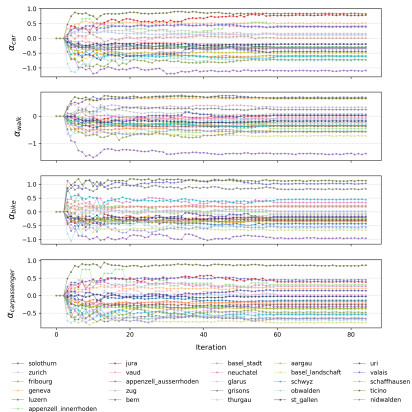
- Switzerland's **26 cantons** differ in taxation, culture, language, geography and infrastructure.
- For this experiment, we calibrate **mode-specific constants for each canton**.
- Can they reproduce each canton's observed mode shares **within one run**?

eqasim Switzerland, 5% population sample.

Targets: Swiss Microcensus mode shares by canton.

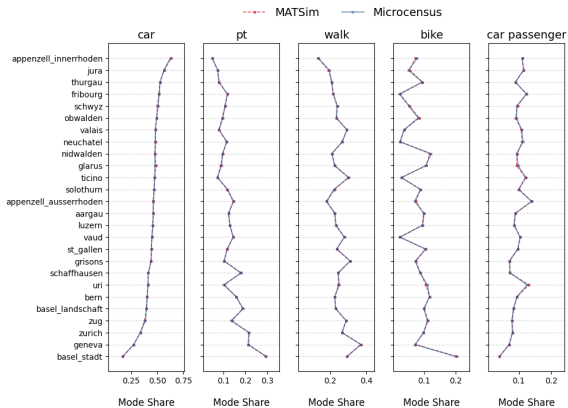


Canton-specific ASCs convergence during the run



Each line is a canton: most ASCs stabilize before iteration 60; some take longer.

One run reproduces mode shares across the 26 cantons



Simulated and observed mode shares agree closely across cantons.

What to watch when calibrating locally

- **Small cantons need larger batches.** Accumulate replanning agents across iterations; we used a threshold of **800 agents** per canton.
- **Update frequency can vary.** Some cantons update less often and take longer to converge.
- **Check the constants as well as the shares.** A close fit can still hide implausible values or overfitting where data are sparse.

Make calibration part of the run

Log choices. Update parameters. Keep running.

- **Canton-specific constants** reproduce local mode shares within a single run.
- Parameters and travel choices adapt together as the simulation progresses.

Code: [eqasim-java](#) | [Python optimizers](#)

Reference: Dib, A. & Balac, M. (2026). *Agent-Based Transport Model Calibration via Online Stochastic Batch Optimization*. Transportation Research Procedia. DOI: 10.3929/ethz-c-000799386

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